

Samuel Tang

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EDUCATION

New York University, Courant – New York, NY

September 2025 – May 2027

Master's in Computer Science

New York University, College of Arts & Science – New York, NY

September 2022 – May 2026

Bachelor of Arts in Computer Science | Minor: Game Design

Dean's List for Academic Year, 2022, 2023

SKILLS

Coding: Python, Java, C++, C, C#, JavaScript, HTML, CSS

Frameworks & Libraries: React.js, Express.js, Flask, Three.js

Technologies: Git, Docker, MongoDB, Linux, VS Code, Blender, Unity, GameMaker Studio, Vite

Languages: English (native), Chinese (fluency at native level)

PROJECTS

PackDex - Pokémon TCG Pack Opening Simulator

- Building a React (Vite) web application that loads live set and card data from the TCGdex API and simulates Pokémon TCG booster pack openings.
- Implementing custom pack-generation logic across multiple game eras, including rarity-weighted pulls, reverse-holo eligibility rules, and set-specific pack structures.
- Creating animated pack reveal sequences and collection views with separate finish tracking for normal, holo, and reverse-holo cards, along with direct TCGplayer search links so users can quickly look up and shop for cards in their collection.
- Designing the system for scalability, supporting future expansion to additional sets, international releases, and market price integration.

BudgetLife - City-Building Budgeting Web App

- Building a full-stack React (Vite) and Node.js mobile application that gamifies personal finance by linking real-world budgeting decisions to the growth and development of a virtual city.
- Implementing an interactive front-end with a pan-and-zoom city map, clickable buildings, and a custom player movement system featuring dynamic pathfinding and obstacle avoidance around environment objects.
- Developing modular React components and context-based state management to support features such as budgeting dashboards, savings goals, transaction history, and progression systems tied to in-game buildings.
- Creating user systems including authentication, account management, and a social layer through a friends system.
- Designing the architecture for scalability, enabling future integration of real financial data, persistent back-end database storage, and expanded gameplay systems such as rewards, city upgrades, and analytics.

Samuel's Room - Portfolio Website

- Built an interactive 3D portfolio environment from scratch, creating all 3D models, UV unwraps, texture maps, and baked light/normal maps in Blender, exported as optimized GLTF/DRACO assets.
- Implemented a custom Three.js WebGL rendering pipeline with custom shaders, lighting setups, mesh instancing, and performance tuning for smooth framerates on desktop and mobile.
- Designed GSAP-driven animation systems to coordinate camera movement, scene transitions, object interactions, and UI choreography into a cohesive experience.
- Integrated Howler.js audio and raycasting-based interactions (modal overlays, playable piano), resolving iOS Safari WebGL issues to keep the experience stable on mobile.

EXPERIENCE

IT Support Technician

June 2025 – August 2025

New York University – New York, NY

- Resolved 60+ individual IT support tickets and assisted colleagues in troubleshooting additional user issues to maintain smooth technical operations.
- Configured and deployed 50+ laptops for NYU College Admissions Leadership Institute students, ensuring proper imaging, user account setup, and system functionality.
- Assisted with moving, unboxing, labeling, and recording large volumes of new computer equipment into the office's asset database in preparation for the upcoming academic year.
- Participated in weekly team meetings to review open tickets and ensure prompt response to all support requests.
- Provided on-site technical support during university events and presentations, including setup assistance and live troubleshooting.

Game Studio Program Fellow

September 2024 – January 2025

EGD Collective – New York, NY

- Collaborated with a smaller specialized team within a larger department to develop *Project Dreamscape*, a game in Unity using C#, while also attending game jams, professional talks, conventions, and more events.
- Worked in a cross-disciplinary team environment with 100+ people that replicates a large AAA studio with an agile development workflow.
- Communicated between teams with scrum, verbal meetings, and written documentation.
- Created scripts for a versatile status effect system and prefab + script for a fireball projectile.